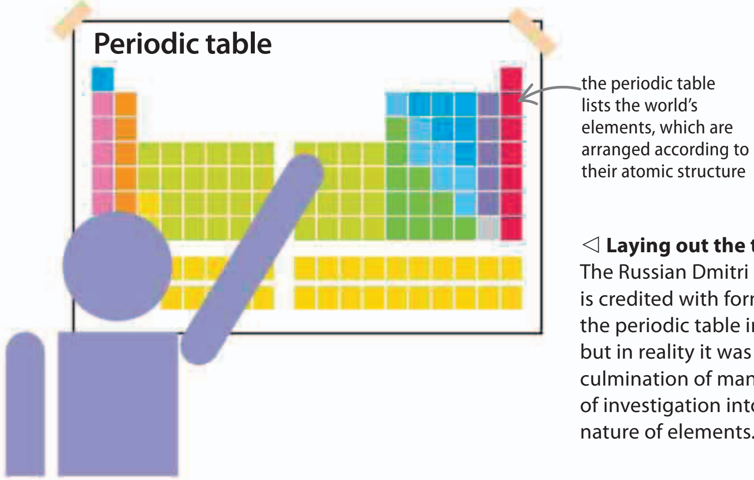
Backing up knowledge

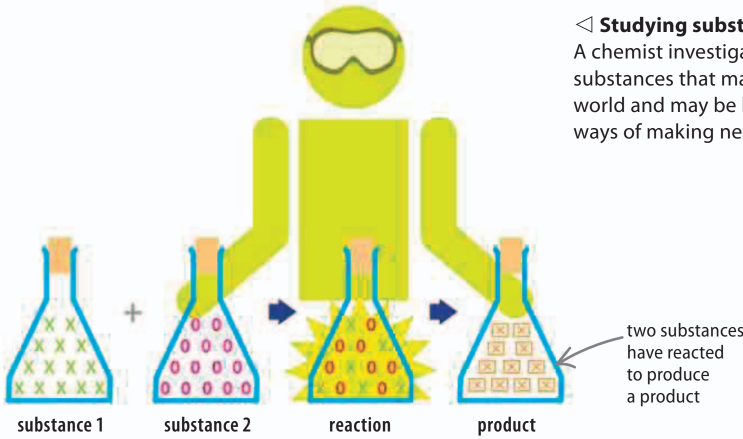
The reason science is such a reliable way of describing nature is because every new piece of knowledge added is only accepted as true if it is based on older pieces of knowledge that everyone already agrees upon. Few scientific breakthroughs are the work of a single mind. When outlining a discovery, scientists always refer to the work of others that they have based their ideas on. In so doing, the development of knowledge can be traced back hundreds, if not thousands, of years.



◁ **Laying out the table**
The Russian Dmitri Mendeleev is credited with formulating the periodic table in 1869, but in reality it was the culmination of many centuries of investigation into the nature of elements.

Specialists

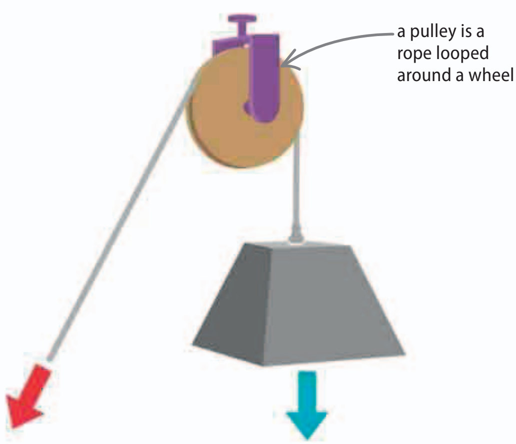
Modern science has been practiced for around 250 years, and in that time great minds have revealed a staggering amount about the nature of life, our planet, and the Universe. Early scientists investigated a wide range of subjects. However, no one alive today can have an expert understanding of all areas of scientific knowledge. There is just too much to know. Instead, scientists specialize in a certain field that interests them, devoting their working lives to unlocking the secrets of that subject.



◁ **Studying substances**
A chemist investigates the substances that make up the world and may be looking for ways of making new ones.

Applying science

Some scientists find explanations for natural phenomena because they are curious—they just like knowing. However, other scientists figure out how the latest understanding of nature might be put to practical use. Applied science and engineering is perhaps the best example of why science is such a powerful tool. If the knowledge discovered by scientists was not correct, none of our high-tech machines would work properly.



◁ **Using force**
Understanding forces and energy explains how it is easier to lift weights with a set of pulleys. For example, lifting a weight with two pulleys requires only half the force needed when using just one.